Section 03

Industrial Security (Bachelor)

IS Security

Section 03 – Industrial Network Architectures

Industrial Network Architectures

Lecture Notes

Department of Computer Science

Outline – Section 03

Learning Objectives

By the end of this section you will be able to

- Apply the Purdue Enterprise Reference Architecture.
- Explain the role of the Industrial DMZ.
- Compare common plant-floor topologies and their trade-offs.
- Apply the IEC 62443 concept of zones and conduits.
- Identify when to use a data diode or unidirectional gateway.

1

Purdue Reference Model

The Purdue Enterprise Reference Architecture

Level 5	Enterprise (ERP, email, web)	
Level 4	Business planning & logistics	FW
Level 3.5	Industrial DMZ (jump host, patch, replicated historian)	FW
Level 3	Site operations (Historian, MES, AV)	
Level 2	Supervisory (SCADA, HMI)	
Level 1	Basic control (PLCs, RTUs, DCS controllers)	
Level 0	Physical process (sensors, actuators)	

Why an Industrial DMZ?

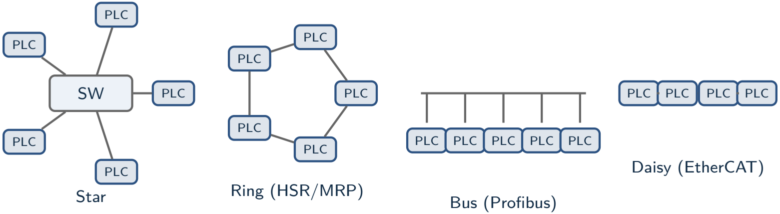
Iron rule

No direct path between IT and OT. Data *lands* in the DMZ; consumers *pull* from there.

2

Topologies

Plant-Floor Topologies



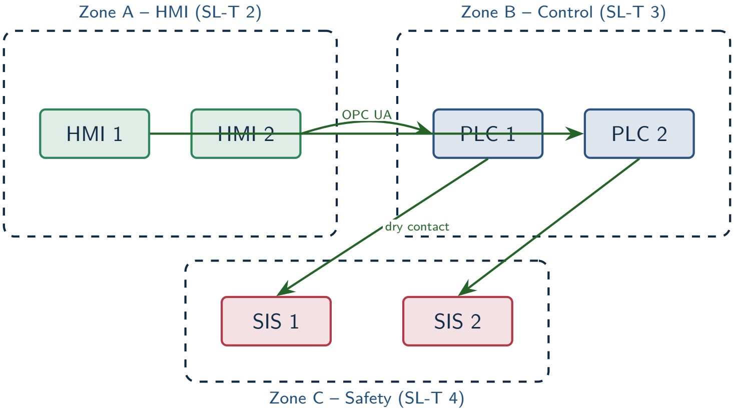
Driver

Topology choice in OT is driven by **determinism** and **availability**, not security.

3

Zones and Conduits

Zones and Conduits (IEC 62443)



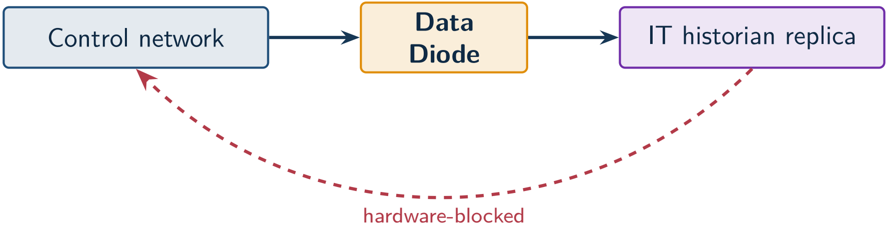
Zones group assets with similar security needs; conduits are the controlled channels between them.

Industrial Firewalls

Modbus function-code allowlist
read coils ✓ write multiple × (after-hours)

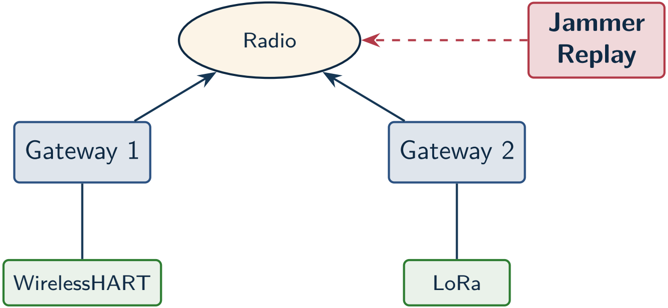
Industrial firewalls do stateful inspection *and* ICS protocol parsing. Often deployed transparently.

Data Diode – One-Way Egress



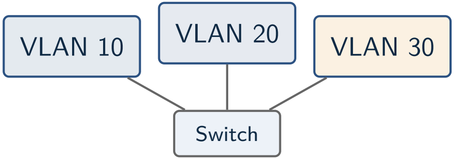
Physical fibre layout enforces the direction. Useful for safety-critical data egress.

Wireless in OT



Convenience vs. attack surface. Never put safety-critical loops over uncontrolled wireless.

VLANs in OT – Useful, Not Sufficient



Caution

VLAN hopping is a real attack A misconfigured trunk port (DTP enabled, native VLAN guessed) lets an attacker hop. Use VLANs for organisation; require a routed firewall for security boundaries.

NAT in OT – When Not to Use

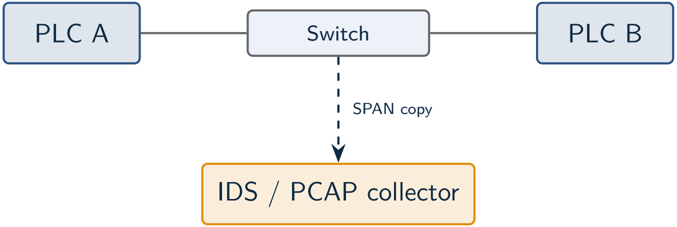
NAT hides, it does not protect

NAT breaks end-to-end addressability – useful for legacy duplicate networks, harmful for incident analysis. Prefer routed, segmented design with explicit firewall rules.

Why it matters for security

Without synchronised clocks, alarm logs and PCAPs are useless in forensics. PTP (IEEE 1588) gives sub-microsecond accuracy on the plant floor; NTP is enough for L3+.

SPAN, TAP, and Visibility



SPAN vs hardware TAP

SPAN can drop packets under load and lets the switch see your collector. A hardware TAP is read-only by construction – preferred for high-fidelity capture.

Wireless Generations on the Plant

Tech	Latency	Range	Typical use
Wi-Fi 6 / 6E	1–10 ms	10–100 m	Mobile HMI, video
WirelessHART	250 ms	100 m	Process sensors
ISA-100.11a	250 ms	100 m	Process sensors
LoRaWAN	seconds	km	Building automation
NB-IoT / LTE-M	seconds	km	Remote telemetry
Private 5G	1 ms	100–500 m	High-density factory

Summary – Section 03

★ Key Takeaway: What to remember from this section

- The Purdue model groups assets into layers; firewalls separate the layers.
- The Industrial DMZ brokers all traffic between corporate IT and the plant.
- Zones and conduits give a structured language for IEC 62443 design.
- Hardware mediation (diodes, gateways) protects the most sensitive flows.

End of Section 03

Industrial Network Architectures

Questions and discussion

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